

Novel Polarimetric Contrast Enhancement Method Based on Minimal Clutter to Signal Ratio Subspace

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Abstract—Enhancing the contrast of target and clutter is a crucial issue in synthetic aperture radar (SAR) image target detection. In this paper, we define a novel subspace, called minimal clutter-to-signal ratio (MCSR) subspace, which can minimize the clutter-to-signal ratio (CSR) by projecting the feature vector to the subspace. Based on MCSR, a novel polarimetric contrast enhancement method is proposed. The MCSR subspace is learned based on the commonly used polarimetric feature vectors extracted from the labeled training SAR image pixels. The feature vectors extracted from candidate SAR image pixels are projected to the MCSR subspace. By calculating the square norm of each transformed feature vector, an enhanced image can be obtained. It is demonstrated that the existing optimization of polarimetric contrast enhancement (OPCE) is a special case of the proposed method to some extent. Experimental results show that our method outperforms the traditional OPCE method on the RadarSat-2 SAR data.

Index Terms—Contrast enhancement, dimension reduction, radar polarimetry, synthetic aperture radar (SAR), target detection.

I. INTRODUCTION

FOR a single-channel synthetic aperture radar (SAR) image, target detection algorithms typically utilize the amplitude information of SAR image to differentiate targets from clutter. However, a polarimetric SAR (PolSAR) image includes four channels from different polarization combinations and provides much rich information than a standard single-channel SAR image. Thus, the PolSAR image can achieve better target detection performance. Exploiting polarimetric data to improve the target detection performance is one of the most important problems in radar polarimetry.

The traditional optimization of polarimetric contrast enhancement (OPCE) is to choose optimal polarization

states for enhancing desired targets versus undesired targets (i.e., clutter). It has been attracting a lot of attention in the past decades [1]–[16]. OPCE falls into two categories: the coherent case and the incoherent case [4]. The coherent OPCE problem is to enhance the single target that can be represented by a Sinclair matrix. The incoherent OPCE problem is to enhance the partial target that is usually represented by a Kennaugh matrix. Efforts have been made solving the OPCE problem [10]–[14]. Yang *et al.* [17] proposed two effective numerical methods to solve the OPCE problem for the coherent and incoherent cases, based on the idea of the Sequential Unconstrained Minimization Technique (SUMT).

OPCE only utilizes intensity relevant features, i.e., Sinclair matrix or Kennaugh matrix. In 2004, Yang *et al.* [18], [19] proposed the generalized OPCE (GOPCE), which not only uses the received power ratio of a desired target and clutter for contrast enhancement but also utilizes the differences in scattering characteristics between the desired target and clutter for enhancement. GOPCE can be divided into two parts: the first part is the same as OPCE, which enhances the SAR image using the intensity relevant features, while the second part enhances the SAR image using the intensity irrelevant features. It is demonstrated in this paper that OPCE can be seen as projecting the feature vector containing the elements of the Sinclair matrix to the 1-D minimal clutter-to-signal ratio (MCSR) subspace with an extra constraint. The dimension of the MCSR subspace and the features that OPCE uses is very limited. Motivated by this, a novel polarimetric contrast enhancement method is proposed in this paper, whose MCSR subspace dimension and exploited features are not limited.

First, we define a novel subspace, called MCSR subspace. Then, the numerical method to find the subspace is derived. In addition, fast solving algorithms for the subspace in some special cases are discussed. The proposed polarimetric contrast enhancement method is implemented based on the MCSR subspace. First, the commonly used intensity relevant polarimetric feature vector for each pixel is extracted from the SAR image. Second, we enhance the contrast of target and clutter by projecting the feature vector to the MCSR subspace. Finally, we calculate the norm of each transformed feature vector and obtain an enhanced image. The experimental results show that our polarimetric contrast enhancement method outperforms the traditional OPCE method.

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This paper is arranged as follows. The definition of the MCSR subspace and the algorithms to find the MCSR subspace are discussed in Section II. The proposed polarimetric enhancing method is presented in Section III. In Section IV, we discuss the relation among proposed polarimetric enhancing method, the traditional OPCE method and the traditional GOPCE method. Section V demonstrates the experimental results and Section VI concludes this paper.

Notation: Throughout this paper, $\mathbb{R}$ and $\mathbb{C}$ denote real and complex domains respectively. Bold-faced upper and lower letters denote matrices and vectors respectively. $\mathbf{O}$ denotes the zero matrix. $\mathbf{I}$ denotes the identity matrix. Superscripts $(\cdot)^T$, $(\cdot)^H$ and $(\cdot)^*$ denote the matrix transpose, the Hermitian transpose and conjugate operators, respectively. $\text{tr}(\cdot)$ denotes the trace operator. $A \setminus B$ denotes set subtraction, i.e., the set containing the elements of A that are not in B . $m - D$ means m -dimensional.

II. MINIMUM CLUTTER TO SIGNAL RATIO SUBSPACE

A. Definition of MCSR Subspace

Denote $\mathbf{t}(T_A)$ and $\mathbf{t}(T_B)$ as the n -D feature vectors of desired target T_A and undesired target (i.e., clutter) T_B , respectively. Then, the m -D clutter-to-signal ratio (CSR) is defined as

$$r = \frac{\mathbb{E}[\|\mathbf{P}_S \mathbf{t}(T_B)\|_2^2]}{\mathbb{E}[\|\mathbf{P}_S \mathbf{t}(T_A)\|_2^2]}, \quad S \in \mathbf{Gr}(m, \mathbb{C}^n) \quad (1)$$

where $\mathbf{Gr}(m, \mathbb{C}^n)$ is the collection of the m -D subspaces of $\mathbb{C}^n$. $\mathbf{P}_S$ is the orthogonal projection matrix of the subspace S . In fact, the m -D CSR is a function defined on a manifold, since $\mathbf{Gr}(m, \mathbb{C}^n)$ is a Grassmann manifold. Denote $\mathbb{E}[\mathbf{t}(T_A)\mathbf{t}^H(T_A)]$ and $\mathbb{E}[\mathbf{t}(T_B)\mathbf{t}^H(T_B)]$ as $\mathbf{R}(T_A)$ and $\mathbf{R}(T_B)$, respectively. Eq. (1) can be equivalently rewritten as

$$\begin{aligned} r &= \frac{\mathbb{E}[\|\mathbf{P}_S \mathbf{t}(T_B)\|_2^2]}{\mathbb{E}[\|\mathbf{P}_S \mathbf{t}(T_A)\|_2^2]} = \frac{\mathbb{E}[\mathbf{t}(T_B)^H \mathbf{P}_S \mathbf{P}_S \mathbf{t}(T_B)]}{\mathbb{E}[\mathbf{t}(T_A)^H \mathbf{P}_S \mathbf{P}_S \mathbf{t}(T_A)]} \\ &= \frac{\mathbb{E}[\text{tr}[\mathbf{P}_S \mathbf{t}(T_B)\mathbf{t}(T_B)^H]]}{\mathbb{E}[\text{tr}[\mathbf{P}_S \mathbf{t}(T_A)\mathbf{t}(T_A)^H]]} = \frac{\text{tr}[\mathbf{P}_S \mathbf{R}(T_B)]}{\text{tr}[\mathbf{P}_S \mathbf{R}(T_A)]}. \end{aligned} \quad (2)$$

Theorem 1: If $\mathbf{R}(T_A) \neq \mathbf{O}$, and there exist no m -D subspace $S \in \mathbf{Gr}(m, \mathbb{C}^n)$ that makes both $\mathbb{E}[\|\mathbf{P}_S \mathbf{t}(T_B)\|_2^2]$ and $\mathbb{E}[\|\mathbf{P}_S \mathbf{t}(T_A)\|_2^2]$ equal zeros, then the m -D CSR r has a minimal value.

Proof: Let

$$f : \mathbf{Gr}(m, \mathbb{C}^n) \setminus Q \rightarrow \mathbb{R},$$

where $Q = \{S | \mathbf{P}_S \mathbf{R}(T_A) = \mathbf{O}\}$ and $f(S) = r = \mathbb{E}[\|\mathbf{P}_S \mathbf{t}(T_B)\|_2^2] / \mathbb{E}[\|\mathbf{P}_S \mathbf{t}(T_A)\|_2^2]$. f is a continuous function.

Since $\mathbf{R}(T_A) \neq \mathbf{O}$, there must exist a subspace $S_0 \in \mathbf{Gr}(m, \mathbb{C}^n)$ that is not an element of Q . Suppose that $f(S_0) = r_0$.

For $S_\alpha \in Q$, when $S \rightarrow S_\alpha$, $f(S) \rightarrow +\infty$. Hence, for S_α , there exists a neighborhood $U(S_\alpha, \varepsilon_\alpha)$ satisfying $f(S) > r_0$ for $S \in U(S_\alpha, \varepsilon_\alpha) \setminus Q$.

Let $U_Q = \cup_{S_\alpha \in Q} U(S_\alpha, \varepsilon_\alpha)$. U_Q is an open set. Therefore $\overline{U_Q}$, the complementary set of U_Q , is a close set.

It is known that a close subset of a compact set is a compact set. Because $\mathbf{Gr}(m, \mathbb{C}^n)$ is compact, $\overline{U_Q}$ is compact.

Since f is continuous on $\overline{U_Q}$, according to the extreme value theorem, f has a minimal value and a maximum value on $\overline{U_Q}$. Denote the minimal value as $r_{\min}$.

Then, $r_{\min} \leq f(S)$ for $S \in \overline{U_Q}$ and $r_{\min} \leq r_0 < f(S)$ for $S \in U_Q \setminus Q$. Hence $r_{\min}$ is the minimal value of f . ■

Definition 1: If $\mathbf{R}(T_A) \neq \mathbf{O}$, and there exist no m -D subspace $S \in \mathbf{Gr}(m, \mathbb{C}^n)$ that makes both $\mathbb{E}[\|\mathbf{P}_S \mathbf{t}(T_B)\|_2^2]$ and $\mathbb{E}[\|\mathbf{P}_S \mathbf{t}(T_A)\|_2^2]$ equal zeros, the minimal value (denoted as $r_{\min}$) of r in (1) is defined as the m -D MCSR. The m -D subspaces corresponding to $r_{\min}$ are m -D MCSR subspaces.

Note that there may exist more than one m -D MCSR subspace corresponding to $r_{\min}$. One extreme example is the case that $\mathbf{R}(T_B)$ is zero matrix. In this case, every m -D subspace of $\mathbb{C}^n$ is an MCSR subspace. For convenience, in the remains of this section, we hypothesize that $\mathbf{R}(T_A) \neq \mathbf{O}$, and there exist no m -D subspace $S \in \mathbf{Gr}(m, \mathbb{C}^n)$ that makes both $\mathbb{E}[\|\mathbf{P}_S \mathbf{t}(T_B)\|_2^2]$ and $\mathbb{E}[\|\mathbf{P}_S \mathbf{t}(T_A)\|_2^2]$ equal zeros.

B. Solutions of MCSR and MCSR Subspace

For convenience, we denote $\mathbf{t}(T_A)$, $\mathbf{t}(T_B)$, $\mathbf{R}(T_A)$, and $\mathbf{R}(T_B)$ as $\mathbf{t}_T$, $\mathbf{t}_C$, $\mathbf{R}_T$, and $\mathbf{R}_C$, respectively.

Finding the m -D MCSR subspaces is equivalent to finding their corresponding orthogonal projection matrix. Then, the problem becomes finding the m -D MCSR $r_{\min}$ and the orthogonal projection matrix of the corresponding m -D MCSR subspace that satisfy

$$r_{\min} = \min_{\dim(\mathbf{P})=m} \frac{\text{tr}(\mathbf{P} \mathbf{R}_C)}{\text{tr}(\mathbf{P} \mathbf{R}_T)}. \quad (3)$$

Since $\mathbf{P}$ is an orthogonal projection matrix, there exists a matrix $\mathbf{U}$ satisfying

$$\begin{aligned} \mathbf{P} &= \mathbf{U} \mathbf{U}^H \\ \mathbf{U}^H \mathbf{U} &= \mathbf{I}. \end{aligned} \quad (4)$$

Then, we have

$$\frac{\text{tr}(\mathbf{P} \mathbf{R}_C)}{\text{tr}(\mathbf{P} \mathbf{R}_T)} = \frac{\text{tr}(\mathbf{U} \mathbf{U}^H \mathbf{R}_C)}{\text{tr}(\mathbf{U} \mathbf{U}^H \mathbf{R}_T)} = \frac{\text{tr}(\mathbf{U}^H \mathbf{R}_C \mathbf{U})}{\text{tr}(\mathbf{U}^H \mathbf{R}_T \mathbf{U})}. \quad (5)$$

Problem (3) can be rewritten as

$$\begin{aligned} r_{\min} &= \min_{\mathbf{U}} \frac{\text{tr}(\mathbf{U}^H \mathbf{R}_C \mathbf{U})}{\text{tr}(\mathbf{U}^H \mathbf{R}_T \mathbf{U})} \\ \text{s.t. } &\mathbf{U}^H \mathbf{U} = \mathbf{I}. \end{aligned} \quad (6)$$

We use the method of Lagrange multipliers to solve (6). Let

$$r(\mathbf{U}) = \frac{\text{tr}(\mathbf{U}^H \mathbf{R}_C \mathbf{U})}{\text{tr}(\mathbf{U}^H \mathbf{R}_T \mathbf{U})} \quad (7)$$

and

$$J(\mathbf{U}) = r(\mathbf{U}) - \sum_{i=1}^m \sum_{j=1}^m \lambda_{ij} \left(\sum_{k=1}^n u_{ki} u_{kj} - \delta_{ij} \right) \quad (8)$$

where λ_{ij} is Lagrange multiplier; u_{kj} is the k th row and j th column entry of $\mathbf{U}$; and δ_{ij} is

$$\delta_{ij} = \begin{cases} 1, & \text{if } i = j \\ 0, & \text{if } i \neq j. \end{cases} \quad (9)$$

Let $(\partial J(U)/\partial U) = 0$. We have

$$\frac{2(\mathbf{R}_C - r\mathbf{R}_T)\mathbf{U}}{\text{tr}(\mathbf{U}^H \mathbf{R}_T \mathbf{U})} - \mathbf{U}(\mathbf{\Lambda}^H + \mathbf{\Lambda}) = 0 \quad (10)$$

where $\mathbf{\Lambda}_{ij} = \lambda_{ij}$. Denote

$$\mathbf{\Lambda}_0 = \frac{(\mathbf{\Lambda}^H + \mathbf{\Lambda})\text{tr}(\mathbf{U}^H \mathbf{R}_C \mathbf{U})}{2}. \quad (11)$$

Equation (10) then becomes

$$(\mathbf{R}_C - r\mathbf{R}_T)\mathbf{U} = \mathbf{U}\mathbf{\Lambda}_0. \quad (12)$$

Since $\mathbf{\Lambda}_0$ is conjugate symmetric, it is unitarily diagonalizable, i.e., there is

$$\begin{aligned} \mathbf{\Lambda}_0 &= \mathbf{T}\mathbf{\Sigma}\mathbf{T}^H \\ \mathbf{\Sigma} &= \text{diag}([\sigma_1, \sigma_2, \dots, \sigma_m]) \end{aligned} \quad (13)$$

where $\mathbf{T}$ is a unitary matrix; and σ_i , $i = 1, 2, \dots, m$, is the i th eigenvalues of $\mathbf{\Lambda}_0$. Equation (12) becomes

$$(\mathbf{R}_C - r\mathbf{R}_T)\mathbf{U}\mathbf{T} = \mathbf{U}\mathbf{T}\mathbf{\Sigma}. \quad (14)$$

Denote $\mathbf{V} = \mathbf{U}\mathbf{T}$. We have

$$(\mathbf{R}_C - r\mathbf{R}_T)\mathbf{V} = \mathbf{V}\mathbf{\Sigma} \quad (15)$$

Due to the fact that

$$\mathbf{P} = \mathbf{U}\mathbf{U}^H = \mathbf{U}\mathbf{T}\mathbf{T}^H\mathbf{U} = \mathbf{U}\mathbf{T}(\mathbf{U}\mathbf{T})^H = \mathbf{V}\mathbf{V}^H \quad (16)$$

all we have to do is to find $\mathbf{V}$. We left multiply both sides of (15) with $\mathbf{V}^H$, trace both sides, and substitute (7) into it. We have

$$\text{tr}(\mathbf{\Sigma}) = 0. \quad (17)$$

Combining (15) and (17), we have the following equations:

$$\begin{cases} (\mathbf{R}_C - r\mathbf{R}_T)\mathbf{V} = \mathbf{V}\mathbf{\Sigma} \\ \text{tr}(\mathbf{\Sigma}) = 0. \end{cases} \quad (18)$$

The MCSR $r_{\min}$ and its corresponding MCSR subspaces must satisfy (18). By observing (18), it can be found that when r is the MCSR, the diagonal elements of $\mathbf{\Sigma}$ (i.e., σ_i , $i = 1, 2, \dots, m$) must be the first m smallest eigenvalues of $\mathbf{R}_C - r_{\min}\mathbf{R}_T$. If not, there must exist an r smaller than $r_{\min}$ satisfying (18).

Let $\sigma_i(r)$, $i = 1, 2, \dots, m$, be the first m smallest eigenvalues of $\mathbf{R}_C - r\mathbf{R}_T$, and $g(r) = \sum_{i=1}^m \sigma_i(r)$. Then, (18) becomes $g(r) = 0$, which can be solved numerically. The quasi-Newton method is used in this paper, and the iterative formula is

$$r^{(k+1)} = r^{(k)} - \frac{g^2(r^{(k)})}{g(r^{(k)}) - g(r^{(k)} - g(r^{(k)}))}. \quad (19)$$

$r^{(0)}$ is set as 0, since $r \geq 0$. The smallest solution is the MCSR $r_{\min}$. The next step is to find the MCSR subspace corresponding to $r_{\min}$. Perform eigenvalue decomposition on $\mathbf{R}_C - r_{\min}\mathbf{R}_T$, and we have

$$\begin{aligned} \mathbf{R}_C - r_{\min}\mathbf{R}_T &= \mathbf{V} \text{diag}([\sigma_1, \sigma_2, \dots, \sigma_n]) \mathbf{V}^H \\ \mathbf{V} &= [\mathbf{v}_1 \quad \mathbf{v}_2 \quad \dots \quad \mathbf{v}_n] \end{aligned} \quad (20)$$

where $\sigma_1 \leq \sigma_2 \leq \dots \leq \sigma_n$. If $\sigma_m \neq \sigma_{m+1}$, $\mathbf{P}_{\min}$ has a unique solution

$$\begin{aligned} \mathbf{V}_{1:m} &= [\mathbf{v}_1 \quad \mathbf{v}_2 \quad \dots \quad \mathbf{v}_m] \\ \mathbf{P}_{\min} &= \mathbf{V}_{1:m} \mathbf{V}_{1:m}^H \\ \mathbf{S}_{\min} &= \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}. \end{aligned} \quad (21)$$

If $\sigma_m = \sigma_{m+1}$, $\mathbf{P}_{\min}$ has multiple solutions. Suppose $\sigma_k = \sigma_m$, $k = m_1, m_1 + 1, \dots, m_2$, where $m_1 \leq m < m_2$. Then, the eigenvectors $\mathbf{v}_{m_1}, \dots, \mathbf{v}_{m_2}$ in (21) can be replaced by any standard orthogonal base of $\text{span}\{\mathbf{v}_{m_1}, \dots, \mathbf{v}_{m_2}\}$ (i.e., the eigenspace corresponding to σ_m). Hence, $\mathbf{P}_{\min}$ has multiple solutions which can be expressed as

$$\begin{aligned} \mathbf{P}_{\min} &= \mathbf{P}_{W_1} + \mathbf{P}_{W_2} \\ W_1 &= \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{m_1-1}\} \\ W_2 &\subset \text{span}\{\mathbf{v}_{m_1}, \mathbf{v}_{m_1+1}, \dots, \mathbf{v}_{m_2}\} \end{aligned} \quad (22)$$

where $\mathbf{P}_{W_1}$ and $\mathbf{P}_{W_2}$ are the orthogonal projection matrices of the subspaces W_1 and W_2 , respectively. The method to find the m -D MCSR and the corresponding m -D MCSR subspace can be summarized as Algorithm 1.

C. Fast Algorithms to Find MCSR Subspace in Some Special Cases

In some special cases, there exist fast algorithms to find the MCSR subspace. In this section, we discuss fast solutions in four special yet practical cases.

1) $m = 1$ and $\mathbf{R}(T_A)$ Is Nonsingular: When $m = 1$, $\mathbf{P}$ can be written as $\mathbf{P} = \mathbf{v}\mathbf{v}^H$, where $\mathbf{v}$ is a norm-1 vector. Then, there is

$$\begin{aligned} r &= \frac{\text{tr}[\mathbf{P}\mathbf{R}(T_B)]}{\text{tr}[\mathbf{P}\mathbf{R}(T_A)]} \\ &= \frac{\mathbf{v}^H \mathbf{R}(T_B) \mathbf{v}}{\mathbf{v}^H \mathbf{R}(T_A) \mathbf{v}}. \end{aligned} \quad (25)$$

$\mathbf{R}(T_A)$ is positive definite, since $\mathbf{R}(T_A)$ is nonsingular. Then, (25) is the generalized Rayleigh quotation. According to the min-max theorem [20], the MCSR $r_{\min}$ is the smallest general eigenvalue of $\mathbf{R}(T_B)$ with respect to $\mathbf{R}(T_A)$, and the corresponding eigenvector $\mathbf{v}_{\min}$ spans the MCSR subspace. If the smallest eigenvalue $r_{\min}$ has multiplicity more than 1, $\mathbf{v}_{\min}$ is not unique, and $\mathbf{v}_{\min}$ can be chosen as any normalized vector in the eigenspace corresponding to $r_{\min}$.

2) $m = n - 1$ and $\dim(\mathbf{R}(T_A)) \geq 2$: When $m = n - 1$, $\mathbf{P}$ can be written as $\mathbf{P} = \mathbf{I} - \mathbf{v}\mathbf{v}^H$, where $\mathbf{v}$ is a norm-1 vector. Then,

$$\begin{aligned} r &= \frac{\text{tr}[(\mathbf{I} - \mathbf{v}\mathbf{v}^H)\mathbf{R}(T_B)]}{\text{tr}[(\mathbf{I} - \mathbf{v}\mathbf{v}^H)\mathbf{R}(T_A)]} \\ &= \frac{\text{tr}[\mathbf{R}(T_B)] - \mathbf{v}^H \mathbf{R}(T_B) \mathbf{v}}{\text{tr}[\mathbf{R}(T_A)] - \mathbf{v}^H \mathbf{R}(T_A) \mathbf{v}} \\ &= \frac{\mathbf{v}^H \{\text{tr}[\mathbf{R}(T_B)] \cdot \mathbf{I} - \mathbf{R}(T_B)\} \mathbf{v}}{\mathbf{v}^H \{\text{tr}[\mathbf{R}(T_A)] \cdot \mathbf{I} - \mathbf{R}(T_A)\} \mathbf{v}}. \end{aligned} \quad (26)$$

Since $\dim(\mathbf{R}(T_A)) \geq 2$, it can be derived that $\text{tr}[\mathbf{R}(T_A)] \cdot \mathbf{I} - \mathbf{R}(T_A)$ is nonsingular. The proof is given in Appendix A. Equation (26) is then a generalized Rayleigh quotation. Hence, the MCSR $r_{\min}$ is the smallest general eigenvalue of $\text{tr}[\mathbf{R}(T_B)] \cdot$

Algorithm 1 Algorithm for Finding m -D MCSR and m -D MCSR Subspace**Input:** $\mathbf{R}_T, \mathbf{R}_C$ **Output:** $\mathbf{P}_{\min}$ Initialization: $r^{(0)} = 0, k = 0$ 1: **repeat**

2:

$$r^{(k+1)} = r^{(k)} - \frac{g^2(r^{(k)})}{g(r^{(k)}) - g(r^{(k)} - g(r^{(k)}))}$$

$$k = k + 1$$

3: **until** $r^{(k)} - r^{(k-1)} < \delta$.4: Let $r_{\min} = r^{(k)}$ 5: Compute the eigenvalues of $\mathbf{R}_C - r_{\min} \mathbf{R}_T$, $\sigma_1 \leq \sigma_2 \leq \dots \leq \sigma_n$, and their corresponding eigenvectors, $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$.6: **if** $\sigma_m \neq \sigma_{m+1}$ **then**

7: There is a unique MCSR subspace. The MCSR subspace and its corresponding orthogonal projection matrix is

$$\mathbf{V} = [\mathbf{v}_1 \ \mathbf{v}_2 \ \dots \ \mathbf{v}_m]$$

$$\mathbf{S}_{\min} = \text{range}(\mathbf{V})$$

$$\mathbf{P}_{\min} = \mathbf{V} \mathbf{V}^H. \quad (23)$$

8: **else if** $\sigma_m = \sigma_{m+1}$ **then**9: $\mathbf{P}_{\min}$ has multiple solutions. Suppose $\sigma_k = \sigma_m$, $k = m_1, m_1 + 1, \dots, m_2$, where $m_1 \leq m < m_2$. Then the MCSR subspaces and their corresponding orthogonal projection matrices are

$$\mathbf{W}_1 = \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{m_1-1}\}$$

$$\mathbf{W}_2 \subset \text{span}\{\mathbf{v}_{m_1}, \mathbf{v}_{m_1+1}, \dots, \mathbf{v}_{m_2}\}$$

$$\mathbf{S}_{\min} = \mathbf{W}_1 \oplus \mathbf{W}_2$$

$$\mathbf{P}_{\min} = \mathbf{P}_{\mathbf{W}_1} + \mathbf{P}_{\mathbf{W}_2} \quad (24)$$

where $\mathbf{P}_{\mathbf{W}_1}, \mathbf{P}_{\mathbf{W}_2}$ is the orthogonal projection matrices of the subspaces $\mathbf{W}_1$ and $\mathbf{W}_2$, respectively.10: **end if**11: **return** $\mathbf{P}_{\min}$

$\mathbf{I} - \mathbf{R}(T_B)$ to $\text{tr}[\mathbf{R}(T_A)] \cdot \mathbf{I} - \mathbf{R}(T_A)$, and the corresponding eigenvector $\mathbf{v}_{\min}$ spans the MCSR subspace. If the smallest eigenvalue $r_{\min}$ has multiplicity more than 1, $\mathbf{v}_{\min}$ is not unique, and $\mathbf{v}_{\min}$ can be chosen as any normalized vector in the eigenspace corresponding to $r_{\min}$.

3) $\mathbf{R}(T_B) = \mathbf{I}$: When $\mathbf{R}(T_B) = \mathbf{I}$, there is

$$r = \frac{\text{tr}[\mathbf{P}\mathbf{R}(T_B)]}{\text{tr}[\mathbf{P}\mathbf{R}(T_A)]} = \frac{\text{tr}[\mathbf{P}]}{\text{tr}[\mathbf{P}\mathbf{R}(T_A)]} = \frac{m}{\text{tr}[\mathbf{P}\mathbf{R}(T_A)]}. \quad (27)$$

Suppose that $\sigma_i, i = 1, \dots, n$ is the eigenvalue of $\mathbf{R}(T_A)$, and $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_n$. Then, $r_{\min} = m / \sum_{i=1}^m \sigma_i$. The MCSR subspace is the space spanned by the eigenvectors corresponding to $\sigma_1, \dots, \sigma_m$. The proof is given in Appendix B. Similar to what has been discussed in Section I-B, the MCSR subspace is unique when $\sigma_m \neq \sigma_{m+1}$, while it has multiple solutions when $\sigma_m = \sigma_{m+1}$. The details will be presented in Appendix B.

It is worth noting that, in this case, projecting the data to the MCSR subspace is equivalent to the well-known principal component analysis (PCA) method.

4) $\mathbf{R}(T_A) = \mathbf{I}$: When $\mathbf{R}(T_A) = \mathbf{I}$, then

$$r = \frac{\text{tr}[\mathbf{P}\mathbf{R}(T_B)]}{\text{tr}[\mathbf{P}\mathbf{R}(T_A)]} = \frac{\text{tr}[\mathbf{P}\mathbf{R}(T_B)]}{\text{tr}(\mathbf{P})} = \frac{1}{m} \text{tr}[\mathbf{P}\mathbf{R}(T_B)]. \quad (28)$$

Suppose that $\sigma_i, i = 1, \dots, n$ is the eigenvalue of $\mathbf{R}(T_B)$, and $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_n$. Then, $r_{\min} = \sum_{i=n-m+1}^n \sigma_i / m$. The MCSR subspace is the space spanned by the eigenvectors corresponding to the first m smallest eigenvalues of $\mathbf{R}(T_B)$. The MCSR subspace is unique when $\sigma_m \neq \sigma_{m+1}$, while it has multiple solutions when $\sigma_m = \sigma_{m+1}$. The proof is similar to the process in Appendix B, and it is suppressed for brevity.

III. PROPOSED POLARIMETRIC ENHANCEMENT METHOD BASED ON MCSR SUBSPACE

In this section, the features used by the proposed method are introduced and the proposed method is demonstrated.

A. Polarimetric Feature Extraction

We extract the following features from a PolSAR image: a 6-D partial scattering vector used in [21]; a 3-D vector containing the three components of Freeman–Durden decomposition [22]–[24]; a 3-D vector containing three components of Pauli decomposition [22]; and a vector containing the three components of Krogager decomposition [25]. Apparently, all these features are relevant to the scattering intensity of the target, and we use all these features to construct a 15-D feature vector denoted as $\mathbf{t}$.

B. Preprocessing of Feature Vectors

Let $\{\mathbf{t}_i(T_A)\}_{i=1}^M$ and $\{\mathbf{t}_i(T_B)\}_{i=1}^N$ be the training set of scattering intensity relevant feature vectors of desired target and undesired target (i.e., clutter), respectively. The self-correlation matrix of T_A and T_B can be estimated as

$$\mathbf{R}(T_A) = \frac{1}{M} \sum_{i=1}^M \mathbf{t}_i(T_A) \mathbf{t}_i(T_A)^H$$

$$\mathbf{R}(T_B) = \frac{1}{M} \sum_{i=1}^M \mathbf{t}_i(T_B) \mathbf{t}_i(T_B)^H. \quad (29)$$

As what we discussed in Section II-A, to guarantee the existence of the MCSR subspace $\mathbf{S}_{\min}$ of $\mathbf{t}(T_A)$ with respect to $\mathbf{t}(T_B)$, two conditions must be satisfied: 1) $\mathbf{R}(T_A) \neq \mathbf{O}$ and 2) there exist no m -D subspace $\mathbf{S} \in \mathbf{Gr}(m, \mathbb{C}^n)$ that makes both $\mathbb{E}[\|\mathbf{P}_{\mathbf{S}} \mathbf{t}(T_B)\|_2^2]$ and $\mathbb{E}[\|\mathbf{P}_{\mathbf{S}} \mathbf{t}(T_A)\|_2^2]$ equal zeros. The first condition is automatically satisfied in practical scenarios.

When the second condition is not satisfied, there exist m -D subspaces that make both $\mathbb{E}[\|\mathbf{P}_{\mathbf{S}} \mathbf{t}(T_B)\|_2^2]$ and $\mathbb{E}[\|\mathbf{P}_{\mathbf{S}} \mathbf{t}(T_A)\|_2^2]$ equal zeros. These subspaces are meaningless since there exists no signal in the subspaces, and it is more important that Algorithm 1 is not applicable in such circumstance. Hence, a preprocessing procedure must be performed on the feature vectors. In particular, we perform a linear transformation to exclude the subspaces where there both T_A and T_B have zero power.

First, perform eigenvalue decomposition on $\mathbf{R}(T_A)$ and suppose $\mathbf{v}_1^A, \mathbf{v}_2^A, \dots, \mathbf{v}_{K_1}^A$ are the eigenvectors corresponding to the eigenvalue 0. Let

$$\mathbf{V}_0^A = [\mathbf{v}_1^A, \mathbf{v}_2^A, \dots, \mathbf{v}_{K_1}^A] \quad (30)$$

and

$$\mathbf{R}'(T_B) = \mathbf{V}_0^{A^H} \mathbf{R}(T_B) \mathbf{V}_0^A. \quad (31)$$

Perform eigenvalue decomposition on $\mathbf{R}'(T_B)$ and suppose $\mathbf{v}_1^B, \mathbf{v}_2^B, \dots, \mathbf{v}_{K_2}^B$ are the eigenvectors corresponding to the eigenvalue 0. Let

$$\mathbf{V}_0^B = [\mathbf{v}_1^B, \mathbf{v}_2^B, \dots, \mathbf{v}_{K_2}^B] \quad (32)$$

where $K_2 \leq K_1$. Then subspace $\text{span}(\mathbf{V}_0^A \mathbf{V}_0^B)$ is exactly the subspace where both T_A and T_B have zero power. The proof is quite direct, and it is omitted here for brevity.

If $\text{span}(\mathbf{V}_0^A \mathbf{V}_0^B) = \emptyset$, there is no subspace in which both T_A and T_B have zero power and there is no need to perform any preprocessing.

When $\dim(\text{span}(\mathbf{V}_0^A \mathbf{V}_0^B)) \geq m$, there exist m -D subspaces in which both T_A and T_B have zero power.

When $\dim(\text{span}(\mathbf{V}_0^A \mathbf{V}_0^B)) < m$, there exist no m -D subspaces in which both T_A and T_B have zero power. Algorithm 1 can still be exploited to find the m -D MCSR subspace of T_A to T_B in such case. However, as we discussed above, $\text{span}(\mathbf{V}_0^A \mathbf{V}_0^B)$ contains no signal, hence we still perform a preprocessing procedure in such case.

Second, find the basis of the orthogonal complement subspace of $\text{span}(\mathbf{V}_0^A \mathbf{V}_0^B)$, $\mathbf{v}_1^C, \mathbf{v}_2^C, \dots, \mathbf{v}_{n-K_2}^C$. One way to find the basis is to calculate the eigenvector corresponding to the 0 eigenvalue of $(\mathbf{V}_0^A \mathbf{V}_0^B)(\mathbf{V}_0^A \mathbf{V}_0^B)^H$. Let

$$\mathbf{V}_0^C = [\mathbf{v}_1^C, \mathbf{v}_2^C, \dots, \mathbf{v}_{K_2}^C]. \quad (33)$$

Then, the feature vectors are preprocessed by

$$\mathbf{t}' = \mathbf{V}_0^{C^H} \mathbf{t}. \quad (34)$$

It is worth noting that, in practical scenario, $\mathbf{R}(T_A)$ and $\mathbf{R}'(T_B)$ usually do not have 0 eigenvalue because the existence of noise. The relatively small eigenvalues are treated as 0 eigenvalue, and in the obtained subspace $\text{span}(\mathbf{V}_0^A \mathbf{V}_0^B)$, there only exists meaningless noise.

C. Projecting the Feature Vectors Onto the MCSR Subspace

In this section, we assume that all feature vectors are preprocessed by the method discussed in Section III-B. Let $\{\mathbf{t}_i(T_A)\}_{i=1}^M$ and $\{\mathbf{t}_i(T_B)\}_{i=1}^N$ be the training sets of the scattering intensity relevant feature vectors of desired target and undesired target (i.e., clutter), respectively. The self-correlation matrices of T_A and T_B can be estimated, respectively, as

$$\begin{aligned} \mathbf{R}(T_A) &= \frac{1}{M} \sum_{i=1}^M \mathbf{t}_i(T_A) \mathbf{t}_i(T_A)^H \\ \mathbf{R}(T_B) &= \frac{1}{M} \sum_{i=1}^M \mathbf{t}_i(T_B) \mathbf{t}_i(T_B)^H. \end{aligned} \quad (35)$$

We find the m -D MCSR subspace $S_{\min}$ of $\mathbf{t}(T_A)$ with respect to $\mathbf{t}(T_B)$ by using Algorithm 1.

In the case that there is a unique MCSR subspace $S_{\min}$, $S_{\min}$ is exactly what we want. In this case, the optimal orthogonal projection matrix is $\mathbf{P}_{\text{opt}} = \mathbf{P}_{\min}$, where $\mathbf{P}_{\min}$ is the orthogonal projection matrix of $S_{\min}$.

In the case that there are more than one MCSR subspace, we select the one that preserves the most power of the target. In particular, suppose that the MCSR subspaces can be represented by

$$\begin{aligned} W_1 &= \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{m_1-1}\} \\ W &= \text{span}\{\mathbf{v}_{m_1}, \mathbf{v}_{m_1+1}, \dots, \mathbf{v}_{m_2}\} \\ W_2 &\subset W \\ S_{\min} &= W_1 \oplus W_2 \\ \mathbf{P}_{\min} &= \mathbf{P}_{W_1} + \mathbf{P}_{W_2}. \end{aligned} \quad (36)$$

Then, the power of T_A in a MCSR subspace is

$$\begin{aligned} P(T_A) &= \|\mathbf{P}_{\min} \mathbf{t}(T_A)\|_2^2 \\ &= \|\mathbf{P}_{W_1} \mathbf{t}(T_A)\|_2^2 + \|\mathbf{P}_{W_2} \mathbf{t}(T_A)\|_2^2 \\ &= \text{tr}(\mathbf{P}_{W_1} \mathbf{R}(T_A)) + \text{tr}(\mathbf{P}_{W_2} \mathbf{R}(T_A)). \end{aligned} \quad (37)$$

Finding $S_{\min}^0 = \max_{S_{\min}} P(T_A)$ is equivalent to finding

$$W_{2,\max} = \max_{W_2 \subset W} (\text{tr}(\mathbf{P}_{W_2} \mathbf{R}(T_A))). \quad (38)$$

We first project $\mathbf{t}(T_A)$ to the subspace W_2 , as given by

$$\begin{aligned} \mathbf{t}_{W_2}(T_A) &= \mathbf{V}_{W_2}^H \mathbf{t}(T_A) \\ \mathbf{V}_{W_2} &= [\mathbf{v}_{m_1}, \mathbf{v}_{m_1+1}, \dots, \mathbf{v}_{m_2}] \end{aligned} \quad (39)$$

where $\mathbf{t}_{W_2}$ is the coordinate of the projected vector, with the columns of $\mathbf{V}_{W_2}$ as the basis of W_2 . Let

$$\begin{aligned} \mathbf{R}_{W_2}(T_A) &= \mathbf{E}[\mathbf{t}_{W_2}(T_A) \mathbf{t}_{W_2}(T_A)^H] \\ &= \mathbf{V}_{W_2}^H \mathbf{R}(T_A) \mathbf{V}_{W_2}. \end{aligned} \quad (40)$$

Then, the problem in (40) is equivalent to find

$$E_{\max} = \max_{\dim(E)=m-m_1+1} \text{tr}(\mathbf{P}_E \mathbf{R}_{W_2}(T_A)). \quad (41)$$

According to the conclusion in Appendix B, $E_{\max}$ is the space spanned by the eigenvectors $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_{m-m_1+1}$ corresponding to the first $m - m_1 + 1$ largest eigenvectors. Then,

$$\begin{aligned} W_2 &= \text{span}\{\mathbf{V}_{W_2} \mathbf{u}_1, \mathbf{V}_{W_2} \mathbf{u}_2, \dots, \mathbf{V}_{W_2} \mathbf{u}_{m-m_1+1}\} \\ S_{\min}^0 &= W_1 \oplus W_2 \\ &= \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_{m_1-1}, \mathbf{V}_{W_2} \mathbf{u}_1, \dots, \mathbf{V}_{W_2} \mathbf{u}_{m-m_1+1}\}. \end{aligned} \quad (42)$$

The orthogonal projection matrix of $S_{\min}^0$ is

$$\begin{aligned} \mathbf{P}_{\min}^0 &= \mathbf{V}_{\min}^0 \mathbf{V}_{\min}^{0^H} \\ \mathbf{V}_{\min}^0 &= [\mathbf{v}_1, \dots, \mathbf{v}_{m_1-1}, \mathbf{V}_{W_2} \mathbf{u}_1, \dots, \mathbf{V}_{W_2} \mathbf{u}_{m-m_1+1}] \end{aligned} \quad (43)$$

and the optimal orthogonal projection matrix is $\mathbf{P}_{\text{opt}} = \mathbf{P}_{\min}^0$. Finally, the SAR image is enhanced by the following formula:

$$P_0 = \|\mathbf{P}_{\text{opt}} \mathbf{t}\|_2^2. \quad (44)$$

As to the choice of the dimension of the MCSR subspace, it should note that the larger the dimension of the MCSR subspace is, the more power of the desired target and undesired

target is contained in the subspace. We prefer the subspace to have a larger enough dimension so that the power of the desired target is preserved. However, we also prefer the subspace to have a small enough dimension to block out the power of the undesired target. The optimal dimension is relevant to the application and the probability distribution of $\mathbf{t}(T_A)$ and $\mathbf{t}(T_B)$. For a target detection task, if there are enough training data, the optimal dimension can be chosen according to the detecting result on the training data.

The proposed polarimetric enhancement method can be summarized as follows.

Algorithm 2 Polarimetric Enhancement Method Based on MCSR Subspace

- 1: Extract the 15-D intensity relevant polarimetric features of each pixel $\mathbf{t}$.
- 2: Find the m -D MCSR subspace of T_B with respect to T_A using the training data $\{\mathbf{t}_i(T_A)\}_{i=1}^M$ and $\{\mathbf{t}_i(T_B)\}_{i=1}^N$ by using Algorithm 1.
- 3: If the MCSR subspace $S_{\min}$ is unique. Then $\mathbf{P}_{\text{opt}} = \mathbf{P}_{\min}$, where $\mathbf{P}_{\min}$ is the orthogonal Projection matrix of $S_{\min}$. If $S_{\min}$ has multiple solutions, find $\mathbf{P}_{\text{opt}}$ following (36) to (43).
- 4: Enhancing the image by

$$P_0 = \|\mathbf{P}_{\text{opt}} \mathbf{t}\|_2^2.$$

IV. RELATIONS OF DIFFERENT POLARIMETRIC CONTRAST ENHANCEMENT METHODS

A. Relation Between the Proposed Method and OPCE

1) *Coherent Case*: In the coherent case, let $\mathbf{S}(T_A)$ and $\mathbf{S}(T_B)$ be the scattering matrices of the desired and undesired target, respectively. Then, the ratio of the echo powers of the two targets is

$$r = \frac{|\mathbf{h}_r^T \mathbf{S}(T_A) \mathbf{h}_t|^2}{|\mathbf{h}_r^T \mathbf{S}(T_B) \mathbf{h}_t|^2} \quad (45)$$

where $\mathbf{h}_t$ and $\mathbf{h}_r$ are the polarization states of the transmission antenna and receive antenna, respectively. The aim of optimal contrast enhancement is to find optimal polarization states to maximize the power ratio r , as given by

$$\begin{aligned} \max_{\mathbf{h}_t, \mathbf{h}_r} & \frac{|\mathbf{h}_r^T \mathbf{S}(T_A) \mathbf{h}_t|^2}{|\mathbf{h}_r^T \mathbf{S}(T_B) \mathbf{h}_t|^2} \\ \text{s.t. } & \mathbf{h}_t^H \mathbf{h}_t = 1 \\ & \mathbf{h}_r^H \mathbf{h}_r = 1. \end{aligned} \quad (46)$$

In order to show the relation between this method and our proposed method, we rewrite (45) into another form. Let

$$\begin{aligned} \mathbf{x} &= [x_1, x_2, x_3, x_4]^T \\ &= (\mathbf{h}_r \otimes \mathbf{h}_t)^* \\ &= [(h_1^r h_1^t)^*, (h_1^r h_2^t)^*, (h_2^r h_1^t)^*, (h_2^r h_2^t)^*]^T \end{aligned} \quad (47)$$

$$\mathbf{s} = \text{vec}(\mathbf{S}) = [S_{11}, S_{12}, S_{21}, S_{22}]^T. \quad (48)$$

Apparently, $\mathbf{x}^H \mathbf{x} = 1$, and $x_1 x_4 = x_2 x_3$. Then (45) can be rewritten as

$$\begin{aligned} r &= \frac{\mathbf{x}^H \mathbf{s}(T_A) \mathbf{s}(T_A)^H \mathbf{x}}{\mathbf{x}^H \mathbf{s}(T_B) \mathbf{s}(T_B)^H \mathbf{x}} \\ &= \frac{\mathbf{x}^H \mathbf{R}_s(T_A) \mathbf{x}}{\mathbf{x}^H \mathbf{R}_s(T_B) \mathbf{x}} \\ &= \frac{\text{tr}[\mathbf{P}_x \mathbf{R}_s(T_A)]}{\text{tr}[\mathbf{P}_x \mathbf{R}_s(T_B)]} \end{aligned} \quad (49)$$

where $\mathbf{R}(T_A) = \mathbf{s}(T_A) \mathbf{s}(T_A)^H$, $\mathbf{R}(T_B) = \mathbf{s}(T_B) \mathbf{s}(T_B)^H$, and $\mathbf{P}_x = \mathbf{x} \mathbf{x}^H$. In fact $\mathbf{P}_x$ is the orthogonal projection matrix of the 1-D subspace spanned by $\mathbf{x}$. Since $x_1 x_4 = x_2 x_3$, we have

$$\begin{aligned} \mathbf{x}^T \mathbf{C} \mathbf{x} &= 0 \\ \mathbf{C} &= \begin{bmatrix} & & & 1 \\ & & -1 & \\ & -1 & & \\ 1 & & & \end{bmatrix}. \end{aligned} \quad (50)$$

The problem (46) then becomes

$$\begin{aligned} \max_{\mathbf{x}} & \frac{\text{tr}[\mathbf{P}_x \mathbf{R}_s(T_A)]}{\text{tr}[\mathbf{P}_x \mathbf{R}_s(T_B)]} \\ \text{s.t. } & \mathbf{x}^T \mathbf{C} \mathbf{x} = 0. \end{aligned} \quad (51)$$

When $\mathbf{R}_s(T_B)$ is a singular matrix, the optimal value of $\mathbf{x}$ may make $\text{tr}[\mathbf{P}_x \mathbf{R}_s(T_B)]$ to be 0. Hence, we transform (51) into its equivalent representation

$$\begin{aligned} \min_{\mathbf{x}} & \frac{\text{tr}[\mathbf{P}_x \mathbf{R}_s(T_B)]}{\text{tr}[\mathbf{P}_x \mathbf{R}_s(T_A)]} \\ \text{s.t. } & \mathbf{x}^T \mathbf{C} \mathbf{x} = 0. \end{aligned} \quad (52)$$

Comparing (52) with (6), we find that $\text{span}(\mathbf{x})$ is the 1-D MCSR subspace. The difference is that (54) has an additional constraint $\mathbf{x}^T \mathbf{C} \mathbf{x} = 0$. The constraint is used to force the enhancing process can be implemented by selecting proper polarimetry state of the transmitting antenna and receiving antenna. The image is finally enhanced by

$$P = \|\mathbf{P}_{x, \text{opt}} \mathbf{s}\|^2. \quad (53)$$

2) *Incoherent Case*: In the incoherent case, $\mathbf{S}(T_A)$ and $\mathbf{S}(T_B)$ change with time or space. Then, the ratio of the echo powers of the two targets is

$$r = \frac{\mathbb{E}[|\mathbf{h}_r^T \mathbf{S}(T_A) \mathbf{h}_t|^2]}{\mathbb{E}[|\mathbf{h}_r^T \mathbf{S}(T_B) \mathbf{h}_t|^2]} \quad (54)$$

which is typically transformed to a representation in terms of Kennan matrices of T_A and T_B , as given by

$$r = \frac{\mathbf{g}_r^T \mathbf{K}(T_A) \mathbf{g}_t}{\mathbf{g}_r^T \mathbf{K}(T_B) \mathbf{g}_t} \quad (55)$$

where $\mathbf{K}(T_A)$ and $\mathbf{K}(T_B)$ are the Kennaugh matrices of T_A and T_B , respectively. $\mathbf{g}_t = [1, g_{t1}, g_{t2}, g_{t3}]^T$ and $\mathbf{g}_r = [1, g_{r1}, g_{r2}, g_{r3}]^T$ are the Stokes vectors of the polarimetry states of the transmitting and the receiving antennas, respectively. Then, the problem becomes

$$\begin{aligned} \max \quad & \frac{\mathbf{g}_r^T \mathbf{K}(T_A) \mathbf{g}_t}{\mathbf{g}_r^T \mathbf{K}(T_B) \mathbf{g}_t} \\ \text{s.t.} \quad & g_{r1}^2 + g_{r2}^2 + g_{r3}^2 = 1 \\ & g_{t1}^2 + g_{t2}^2 + g_{t3}^2 = 1. \end{aligned} \quad (56)$$

The algorithm for solving (56) can be found in [17]. In this paper, we transform (56) to another form to show the relation between OPCE and our method. Following the same process in the coherent case, we have

$$\begin{aligned} \min_x \quad & \frac{\text{trace}[\mathbf{P}_x \mathbf{R}_s(T_B)]}{\text{trace}[\mathbf{P}_x \mathbf{R}_s(T_A)]} \\ \text{s.t.} \quad & \mathbf{x}^T \mathbf{C} \mathbf{x} = 0. \end{aligned} \quad (57)$$

The only difference between (57) and (52) is that $\mathbf{R}(T_A) = \mathbf{E}[s(T_A)s(T_A)^H]$ and $\mathbf{R}(T_B) = \mathbf{E}[s(T_B)s(T_B)^H]$ here. The image is finally enhanced by

$$\mathbf{P} = \|\mathbf{P}_{x,\text{opt}} \mathbf{s}\|^2. \quad (58)$$

In both coherent case and incoherent case, OPCE projects the feature vector containing the elements of the Sinclair matrix to the 1-D MCSR subspace with an extra constraint (i.e., $\mathbf{x}^T \mathbf{C} \mathbf{x} = 0$). Compare to OPCE, the proposed method uses unlimited features. Hence the proposed method is more flexible. To some extent, OPCE can be seen as a special case of our proposed method.

B. Relation Between the Proposed Method and GOPCE

GOPCE is a modified version of OPCE in the incoherent case. GOPCE not only utilizes the scattering intensity relevant features (i.e., Kennaugh matrix) but also scattering intensity irrelevant features. GOPCE uses three scattering intensity irrelevant features: 1) the similarity parameter between the target and a plane; 2) the similarity parameter between the target and a dihedral d_2 ; and 3) the entropy of the target d_3 . The definition of d_1 and d_2 can be referred to [18], and the definition of d_3 can be referred to [22]. Different from scattering intensity relevant features, these three features are not relevant to the scattering intensity, but relevant to the scattering mechanism of the target. Let $\mathbf{d} = [d_1, d_2, d_3]^T$. The aim of GOPCE is to solve the following optimization:

$$\begin{aligned} \min_{\mathbf{g}_t, \mathbf{g}_r} \quad & \frac{\mathbf{g}_r^T \mathbf{K}(T_B) \mathbf{g}_t}{\mathbf{g}_r^T \mathbf{K}(T_A) \mathbf{g}_t} \times \frac{E[|\mathbf{d}(T_B)^T \mathbf{w}|^2]}{E[|\mathbf{d}(T_A)^T \mathbf{w}|^2]} \\ \text{s.t.} \quad & g_{r1}^2 + g_{r2}^2 + g_{r3}^2 = 1 \\ & g_{t1}^2 + g_{t2}^2 + g_{t3}^2 = 1 \\ & w_1^2 + w_2^2 + w_3^2 = 1. \end{aligned} \quad (59)$$

where $\mathbf{w} = [w_1, w_2, w_3]^T$ is a weight vector. The optimization can be divided into two separate optimizations

$$\begin{aligned} \min_{\mathbf{g}_t, \mathbf{g}_r} \quad & \frac{\mathbf{g}_r^T \mathbf{K}(T_B) \mathbf{g}_t}{\mathbf{g}_r^T \mathbf{K}(T_A) \mathbf{g}_t} \\ \text{s.t.} \quad & g_{r1}^2 + g_{r2}^2 + g_{r3}^2 = 1 \\ & g_{t1}^2 + g_{t2}^2 + g_{t3}^2 = 1 \\ \min_{\mathbf{w}} \quad & \frac{E[|\mathbf{d}(T_B)^T \mathbf{w}|^2]}{E[|\mathbf{d}(T_A)^T \mathbf{w}|^2]} \\ \text{s.t.} \quad & w_1^2 + w_2^2 + w_3^2 = 1. \end{aligned} \quad (60)$$

After obtaining the optimal parameters $\mathbf{g}_{t,\text{opt}}$, $\mathbf{g}_{r,\text{opt}}$ and $\mathbf{w}_{\text{opt}}$, the image is enhanced by

$$\begin{aligned} \mathbf{P} &= \mathbf{P}_1 \times \mathbf{P}_2 \\ \mathbf{P}_1 &= \mathbf{g}_{r,\text{opt}}^T \mathbf{K} \mathbf{g}_{t,\text{opt}} \\ \mathbf{P}_2 &= |\mathbf{d}^T \mathbf{w}_{\text{opt}}|^2. \end{aligned} \quad (61)$$

According to what we have discussed in Section IV-A, solving the first optimization in (60) is equivalent to finding the orthogonal projection matrix of the 1-D MCSR subspace, $\mathbf{P}_{x,\text{opt}}$, with an extra constraint (i.e., $\mathbf{x}^T \mathbf{C} \mathbf{x} = 0$). Solving the second optimization in (60) is equivalent to finding the orthogonal projection matrix of the 1-D MCSR subspace, $\mathbf{P}_{w,\text{opt}}$, where $\mathbf{P}_{w,\text{opt}} = \mathbf{w}_{\text{opt}} \mathbf{w}_{\text{opt}}^H$. The image can be equivalently enhanced by

$$\begin{aligned} \mathbf{P} &= \mathbf{P}_1 \times \mathbf{P}_2 \\ \mathbf{P}_1 &= \|\mathbf{P}_{x,\text{opt}} \mathbf{s}\|^2 \\ \mathbf{P}_2 &= \|\mathbf{P}_{w,\text{opt}} \mathbf{d}\|^2. \end{aligned} \quad (62)$$

Basically, GOPCE separately utilizes two types of features (i.e., scattering intensity relevant features and scattering intensity irrelevant features) to enhance the image and obtains two different enhanced images. Then, the two images are fused into one by multiplying the corresponding pixels of the two enhanced images.

V. NUMERICAL EXPERIMENTS

A. Experiments for Validating Algorithm 1

1) *Case When the MCSR Subspace Has Unique Solution:* We simulate target data and clutter data distributed on two different planes in the 3-D real Euclidean space. The data are generated according to the following expressions:

$$\begin{aligned} \mathbf{t}(T_A) &= \varepsilon_1 [1, 0, 0]^T + 0.5 \varepsilon_2 [0, 1, 0]^T \\ \mathbf{t}(T_B) &= \varepsilon_3 \left[\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3} \right]^T + 0.5 \varepsilon_4 [0, 1, 0]^T \\ \varepsilon_1, \varepsilon_2, \varepsilon_3, \varepsilon_4 &\sim N(0, 1) \end{aligned} \quad (63)$$

where $N(0, 1)$ stands for the Gaussian distribution with mean of 0 and variance of 1. We generate 500 samples for T_A and T_B . The data are shown in Fig. 1(a). Then, the data are projected to a plane represented by its norm vector, and the CSR is calculated. The norm vector can be uniquely

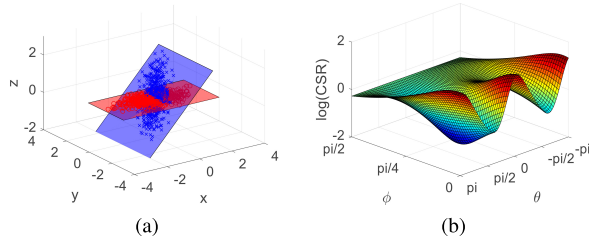


Fig. 1. (a) Target data are marked with red circles and distributed in the red plane. The clutter data are marked with blue crosses and distributed in the blue plane. (b) CSR when projected onto different planes.

TABLE I
MINIMAL CSRS AND NORM VECTOR OF THE MCSR
PLANES FOUND BY DIFFERENT METHODS

	The minimal CSR	Norm vector
Search Method	0.1461	$[0.4570, 0.7028, 0.5452]^T$
Algorithm 1	0.1460	$[0.4823, 0.6947, 0.5337]^T$

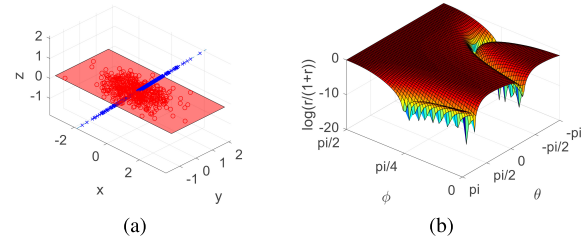


Fig. 2. (a) Target data are marked with red circles and distributed in the red plane. The clutter data are marked with blue crosses and distributed on the blue line. (b) CSR when projected onto different lines.

represented by its azimuthal angle θ and elevation angle φ . The plot of $\log \text{CSR}$ versus θ and φ is shown in Fig. 1(b). It is obvious that the plot has minimal value. The corresponding minimal CSR and norm vector of the MCSR plane are listed in Table I. Then, Algorithm 1 is used to calculate the minimal CSR and the corresponding norm vector which are also listed in Table I. The results obtained by the two methods are very close, which validates the effectiveness of Algorithm 1.

2) *Case When the MCSR Subspace Have Multiple Solutions*: We simulate target data distributed on a plane and clutter data distributing along a line in the 3-D real Euclidean space. The data are generated according to the following expressions:

$$\begin{aligned} \mathbf{t}(T_A) &= \varepsilon_1[1, 0, 0]^T + 0.5\varepsilon_2[0, 1, 0]^T \\ \mathbf{t}(T_B) &= \varepsilon_3 \left[\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3} \right]^T \\ \varepsilon_1, \varepsilon_2, \varepsilon_3 &\sim N(0, 1). \end{aligned} \quad (64)$$

Again, we generate 500 samples for T_A and T_B . The data are shown in Fig. 2(a). Then, the data are projected to the 1-D subspace, i.e., a line represented by its direction vector, and the CSR is calculated. The direction vector can be uniquely represented by its azimuthal angle θ and elevation angle φ . Since the log of CSR r may be negative infinite, we use $r/(1+r)$ instead of r to measure the CSR. The plot of

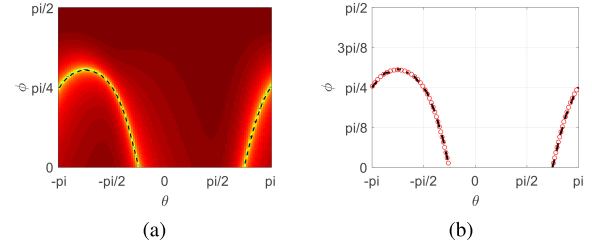


Fig. 3. (a) 2-D image of the CSRs. The MCSR subspaces are marked with black dashed lines. (b) Red line marked with red circle represent the straight lines in the plane. The black dashed line represents the MCSR subspaces shown in (a).

$\log(r/(1+r))$ versus θ and φ is shown in Fig. 2(b). It is obvious that the plot has more than minimal value. The 2-D version of the plot in Fig. 2(b) is shown in Fig. 3(a). The points on the black dashed line have minimal values. The MCSR obtained according to Fig. 2(b) is 3.2437×10^{-8} . Then, we use Algorithm 1 to find the MCSR and the MCSR subspaces. We obtain that the MCSR is 0. According to (21), we have

$$\begin{aligned} \mathbf{R}_C &= \mathbf{V} \mathbf{\Sigma} \mathbf{V}^T \\ \mathbf{\Sigma} &= \text{diag}(0, 0, 528.7) \\ \mathbf{V} &= \begin{bmatrix} 0.4735 & -0.6652 & 0.5574 \\ -0.8128 & -0.0775 & 0.5574 \\ 0.3393 & 0.7427 & 0.5574 \end{bmatrix} \end{aligned} \quad (65)$$

and the MCSR subspaces are obtained by

$$\begin{aligned} S_{\min} &\subset W_2 \\ W_2 &= \text{span} \left\{ \begin{bmatrix} 0.4735 \\ -0.8128 \\ 0.3393 \end{bmatrix}, \begin{bmatrix} -0.6652 \\ -0.0775 \\ 0.7427 \end{bmatrix} \right\}. \end{aligned} \quad (66)$$

In fact, it can be known from (65) and (66) that W_2 is the plane orthogonal to the line along which the clutter data are distributed on. That is, any line in the plane is an MCSR subspace. We represent all the lines on the plane in the (θ, φ) plane, as shown in Fig. 3(b). The points on the red lines marked with red circle represent the straight lines in the plane W_2 , while the points on the black dashed lines represent the MCSR subspaces shown in Fig. 3(a). The red lines coincide with the black lines, which again proves that Algorithm 1 successfully finds the MCSR subspaces.

B. Experiments for Target Detection

In order to provide a validation of the proposed method, a C-band RadarSat-2 Fine-Quad-mode fully PolSAR data set is exploited. The resolutions are about 8 and 12 m in the azimuth and range directions, respectively. The data were acquired on August 4, 2010. The pseudocolor image of the data is shown in Fig. 4(a). Two chips are cut from the full image. The pseudocolor image of one chip is shown in Fig. 4(b) and this chip is used for testing. The pseudocolor image of the other chip is shown in Fig. 4(c) and this chip is used for training. We use the OPCE method, the GOPCE method and the spanned image in comparisons with the proposed methods.

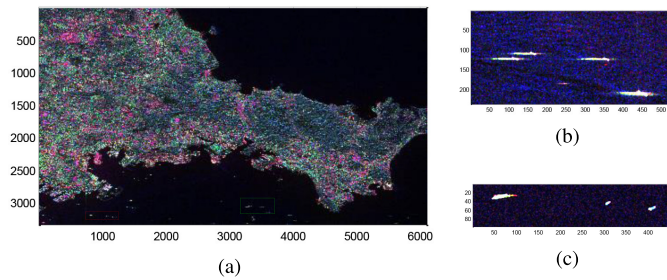


Fig. 4. (a) Pseudocolor image of the RadarSat-2 data. (b) Pseudocolor image of the region in the green window. (c) Pseudocolor image of the region in the red window.

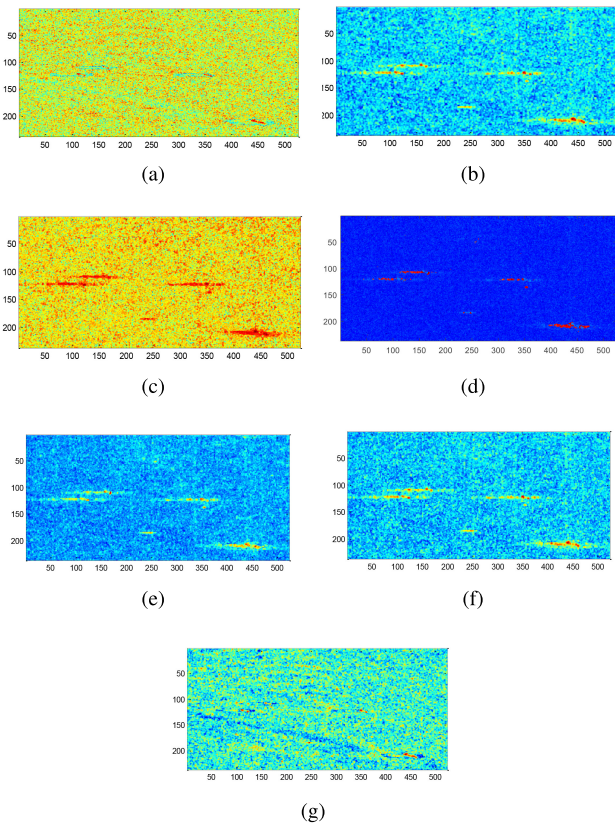


Fig. 5. (a) Log span image of the test data with a 0-dB SCR. (b) Log image enhanced by OPCE method. (c) Log image enhanced by GOPCE method. (d)–(f) Log image enhanced by the proposed method with the dimension of the MCSR subspace of 2, 7, and 10. (g) Log image of the norm of the 15-D feature vector used by the proposed method.

The test image is scaled to have a signal to clutter ratio of 0 dB. The log span image of the test image is shown in Fig. 5(a). The chip shown in Fig. 4(c) is used to train OPCE, GOPCE, and the proposed method. Then, OPCE, GOPCE, and the proposed method are used to enhance the chip in Fig. 4(b). The log images enhanced by the OPCE and GOPCE are shown in Fig. 5(b) and (c), respectively. The log image enhanced by the proposed method with subspace dimensions of 2, 7, and 10 are shown in Fig. 5(d)–(f) respectively. The log image for the norm of the feature vector is shown in Fig. 5(g). In fact, the orthogonal projection matrix of the 15-D MCSR subspace is identical matrix, since the dimension of the feature vector is 15. Hence, the image value enhanced by the proposed

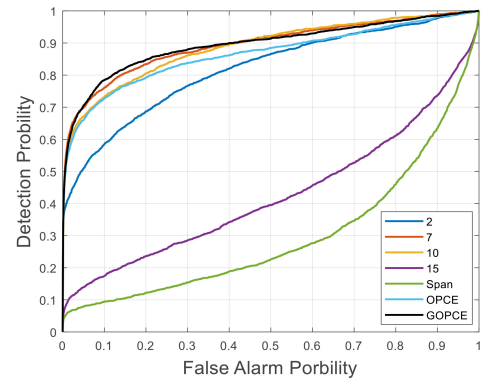


Fig. 6. ROC curves of the image enhanced by the proposed method with the different subspace dimensions and other methods.

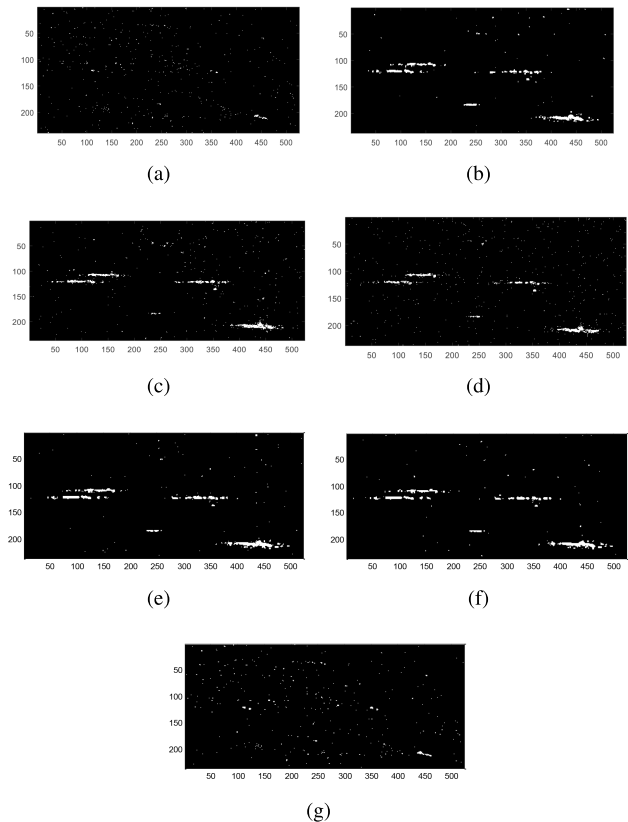


Fig. 7. Detecting results with a false alarm ratio of 0.5% of (a) span image, (b) image enhanced by OPCE method, and (c) image enhanced by GOPCE method. (d)–(f) Image enhanced by the proposed method with the dimension of the MCSR subspace of 2, 7, and 10. (g) Image of the norm of the 15-D feature vector used by the proposed method.

method with 15-D subspace is the case without dimension reduction. Fig. 6 shows the receiver operating characteristic (ROC) curves of the different methods. It is worth noting that the detection rate and the false alarm rate are calculated based on counting pixels. Fig. 7 shows the detection results of the different methods with a false alarm rate of 0.5%.

It can be seen that the targets in Fig. 5(b)–(f) are more obvious than the targets in Fig. 5(a) which validates the proposed method, the OPCE method, and the GOPCE method. The targets in Fig. 5(d)–(f) are more obvious than the targets

in Fig. 5(g), which demonstrates that the proposed method successfully preserves the power of the target and block the power of the clutter. These observations can also be seen in Fig. 6, where the ROC curves corresponding to Fig. 5(b)–(f) are higher than the ROC curves corresponding to Fig. 5(a) and (g).

By comparing the results of the proposed method with 2-D MCSR subspace, and the proposed method with 7-D MCSR subspace, it can be found that the proposed method with 2-D MCSR subspace make targets more obvious but has lower ROC curves. The reason is that when the MCSR subspace dimension is low, pixels that have features close to the clutter are overlooked and only pixels that have features very different from the clutter are enhanced. When the MCSR subspace dimension is high, more pixels of the target are included in the subspace, and more pixels of the target are enhanced. On the other hand, more pixels of the clutter are also included in the MCSR subspace in this case.

If we evaluate the different methods by the ship numbers detected from the image enhanced by the methods, the proposed method and OPCE has the same performance since all the ships are detected from the image enhanced by the two methods. On the other hand, by comparing the ROC curves of the proposed method with 7-D MCSR subspace and OPCE, we find that the proposed method has higher ROC curve. In this aspect, the proposed method outperforms OPCE, if a proper MCSR subspace dimension is chosen.

It is quite complicative to compare the proposed method with GOPCE. It can be seen that the targets in the image enhanced by GOPCE is very obvious, and GOPCE almost has the same detection performance with the proposed method with 7-D the MCSR subspace, if we compare the ROC curves of the two methods. This is because GOPCE utilizes scattering intensity irrelevant features. But it does not mean that GOPCE is necessarily better than the proposed method. There are two proofs.

Now, we discuss the first evidence. We can modify the proposed method by introducing the thought to GOPCE into our proposed method. The image is enhanced by

$$\begin{aligned} P_{\text{modified}} &= P_0 \times P_2 \\ P_0 &= \|P_{\text{opt}}t\|_2^2 \\ P_2 &= \|P_{w,\text{opt}}d\|^2 \end{aligned} \quad (67)$$

where P_0 is the image enhanced by the proposed method, as described in (44), and P_2 is the image enhanced by utilizing scattering intensity irrelevant features as described in (62). Then, the modified proposed method is used to detect the ships. The ROC curve of the modified proposed method with 7-D MCSR subspace is shown in Fig. 8(a). The log image enhanced by the modified proposed method with 7-D MCSR subspace and its detection result are shown in Fig. 9(a) and (b), respectively. It can be seen that the ROC curve of the modified proposed method is higher than that of GOPCE, and the modified proposed method detects more ship pixels than GOPCE. Hence, our proposed method will outperform GOPCE if scattering intensity irrelevant features are utilized by our proposed method.

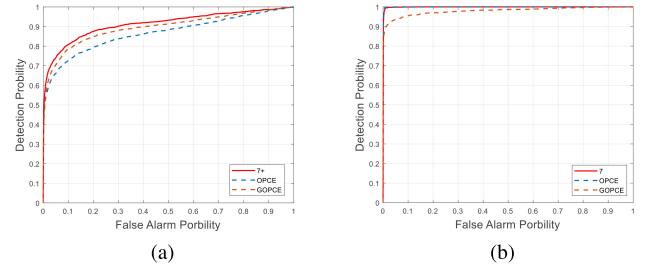


Fig. 8. (a) ROC curves of the image enhanced by the modified proposed method with 7-D MCSR subspace (7+), OPCE, and GOPCE. (b) ROC curves of the image enhanced by the proposed method with 7-D MCSR subspace, OPCE, and GOPCE for a chip with the signal clutter ratio of 15 dB.

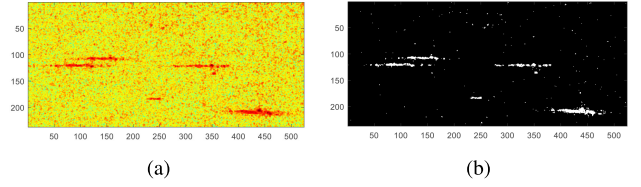


Fig. 9. (a) Log image and enhanced by the modified proposed method with 7-D MCSR subspace. (b) Detection result with a false alarm ratio of 0.5% of the image enhanced by the modified proposed method with 7-D MCSR subspace.

Then, we discuss the second evidence. We change the signal to clutter ratio of the test image. The test image is scaled to have a signal to clutter ratio of 15 dB. Then, the proposed method, OPCE, and GOPCE are used to detect the ships in the test image. The ROC curves of the three methods are shown in Fig. 8(b), and the enhanced image and their detection results are shown in Fig. 10. It can be seen that the proposed method and OPCE have higher ROC curves than GOPCE, and the proposed method and OPCE detect more pixels of the smallest ship than GOPCE. The reason is that the way GOPCE utilizes scattering intensity irrelevant features does not always improve the detection performance of OPCE. Whether GOPCE can outperform OPCE depends on which are the better features, scattering intensity relevant features or scattering intensity irrelevant features. How to combine scattering intensity relevant features and scattering intensity irrelevant features to enhance PolSAR images is another problem which is out of the scope of this paper and needs to be further studied.

The experiments are run on a computer with Intel core i5-4590 CPU at 3.30 GHz, 8-GB memory and software MATLAB (R2017a). Table II presents the time consuming of the proposed method, OPCE, and GOPCE. We divide the entire time-consuming into three parts, the time for feature extraction, the time for solving for parameters, and the time for image enhancing.

For the proposed method, features extraction refers to extracting the 15 scattering intensity relevant features; solving for parameters refers to solving for the orthogonal projection matrix of the MCSR subspace; and image enhancing refers to calculating the enhanced image according to (44). For OPCE, feature extraction refers to calculating the Kennaugh matrix; solving for parameters refers to solving for the optimal

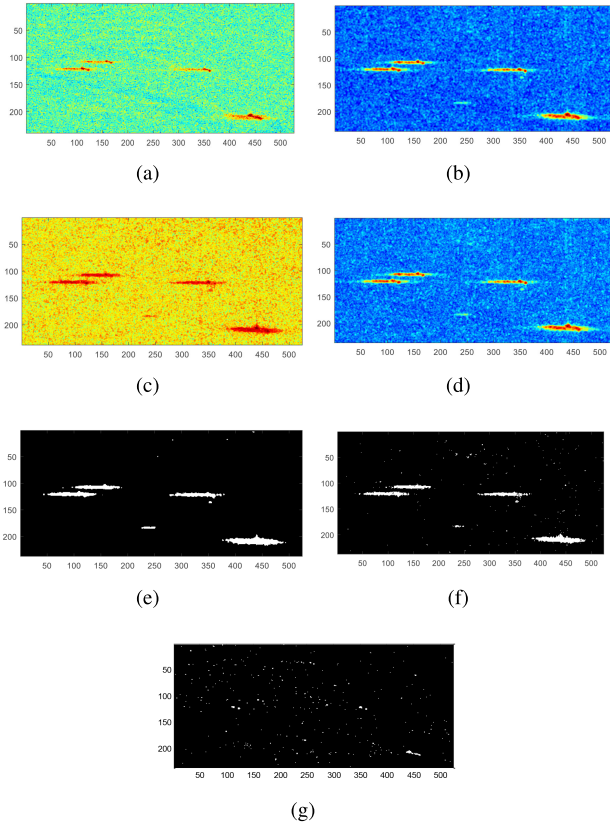


Fig. 10. Log span image of the test data with a 15 dB SCR. (b)–(d) Log image enhanced by OPCE, GOPCE, and the proposed method with 7-D MCSR subspace. (e)–(g) Detection results with a false alarm ratio of 0.5% of the images enhanced by OPCE, GOPCE, and the proposed method with 7-D MCSR subspace.

TABLE II
TIME-CONSUMING OF DIFFERENT METHODS

	Features extraction	Solving for parameters	Image enhancing
OPCE	3.58 s	3.32E-3 s	2.65E-1 s
GOPCE	11.6 s	4.19E-3 s	7.23E-1 s
Proposed method	3.67 s	5.06E-4 s	3.72E-2 s

polarization states of the transmitting and receiving antennas; and image enhancing refers to calculating the receiving power of the image under the optimal polarization states of the antennas. For GOPCE, feature extraction refers to calculating the Kennaugh matrix and the three scattering intensity irrelevant features; solving for parameters refers to solving for the optimal polarization states of the transmit and receive antennas and the optimal weight for the three scattering intensity irrelevant features; and image enhancing refers to calculating the enhanced image according to (62). It can be seen from Table II that features extraction is the most time-consuming part for all the three methods, and the total time-consuming of the proposed methods is relatively low, as compared to OPCE and GOPCE.

C. Experiments for Target Detection in a Larger Scene

To further test our proposed method, a larger test chip contains more than 25 ships is cut from the RadarSat-2 data.

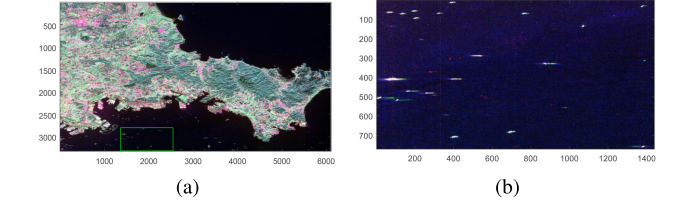


Fig. 11. (a) Pseudocolor image of the RadarSat-2 data. (b) Pseudocolor image of the region in the green window.

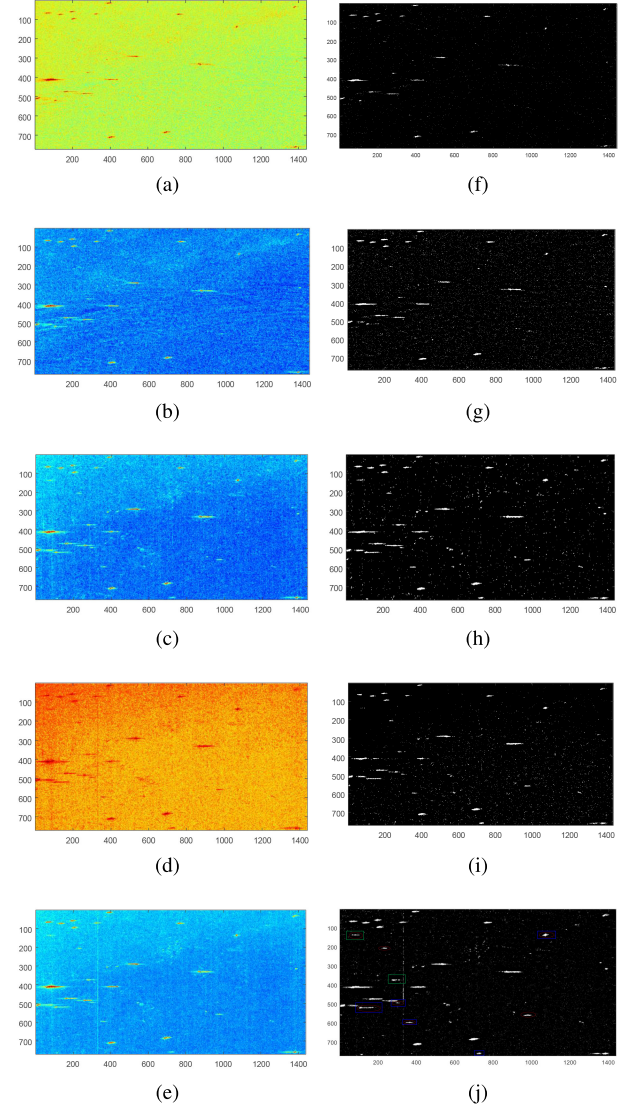


Fig. 12. (a) Log span image. (b) Log image of the norm of the 15-D feature vector of the proposed method. (c)–(e) Log images enhanced by OPCE, GOPCE, and the proposed method with 7-D MCSR subspace. (f)–(j) Detection results corresponding to the images in (a)–(e).

The pseudocolor image of the test chip is shown in Fig. 11(b). The log span image of the test chip is shown in Fig. 12(a) and the log image of the norm of the 15-D feature vector used by the proposed method is shown in Fig. 12(b). The chip in Fig. 4(b) is used as the training data. The proposed method, OPCE, and GOPCE are used to enhance the test chip. Fig. 12(c)–(e) shows the log images enhanced by OPCE, GOPCE, and the proposed method, respectively.

TABLE III
PIXEL NUMBERS OF THE FIRST TEN LARGEST SHIPS DETECTED BY OPCE AND THE PROPOSED METHOD (PM)

Ship No.	1	2	3	4	5	6	7	8	9	10	Total
OPCE	912	660	507	425	385	383	358	343	306	259	4538
PM	934	615	479	470	440	431	409	398	332	309	4817

The sea clutter in images in Fig. 12(a)–(e) is assumed to follow the normal distribution. The constant false alarm rate detector is used to detect the ships in the images. The false alarm rate is set to be 0.3%. The detection results are shown in Fig. 12(j)–(f). In Fig. 12(j), red ellipses mark the ships that are not detected in the log span image, blue rectangles mark the ships that are not detected in log image of the norm of the 15-D feature vector, and green rectangles mark the ships that are not detected in the log image enhanced by GOPCE. All ships are detected from the images enhanced by the proposed method and OPCE.

From the detection results, the following facts can be found. The proposed method and OPCE have the best detection results. The log image enhanced by the proposed method has better detection result than the log image of the norm of the 15-D feature vector, which illuminates that proposed method successfully preserves the power of the target and block the power of the clutter. The log image enhanced by the proposed method, OPCE, and GOPCE have better detection results than the log span image, which proves the validity of the three methods.

The proposed methods and OPCE outperform GOPCE because, as we discussed in Section V-B, the way GOPCE utilizes scattering intensity irrelevant features do not always improve the detection performance of OPCE. Whether GOPCE can outperform OPCE depends on which are the better features, scattering intensity relevant features or scattering intensity irrelevant features. The signal-to-clutter ratio of the test chip is more than 20 dB. In such a situation, the scattering intensity relevant features are the better features and the way GOPCE utilizing scattering intensity irrelevant features will damage the detection performance.

Both the proposed method and OPCE detect all the ships. However, there is some difference between the two methods. If a connected component with a large number of pixels in the detection result is considered as a ship, the pixels of every detected target can be counted. Table III separately presents the pixel numbers of the first ten largest ships detected by OPCE and proposed method. For eight of the ten ships, the proposed method detects more pixels, and for the ten ships, the total number of pixels detected by the proposed method is also greater than the number of the pixels detected by OPCE. On this aspect, the proposed method outperforms the OPCE method.

VI. CONCLUSION

In this paper, we define a novel subspace, called MCSR subspace, which can minimize the CSR by projecting the feature vector to the subspace. Based on which, a novel

polarimetric contrast enhancement method is proposed. It is demonstrated in this paper that OPCE can be seen as special cases of the proposed method to some extent. Compared to OPCE and GOPCE, the proposed method is more flexible, because it uses unlimited features. How to combine scattering intensity relevant features and scattering intensity irrelevant features to enhance PolSAR images needs to be further studied. Experiments on simulation data show the effectiveness of algorithms to find the MCSR subspace. The experiment on measured data show that the proposed method outperforms the traditional OPCE method.

APPENDIX A

In this section, we prove that

$$\text{tr}[\mathbf{R}(T_A)] \cdot \mathbf{I} - \mathbf{R}(T_A)$$

is nonsingular when $\dim(\mathbf{R}(T_A)) \geq 2$. Let $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_n$ be the eigenvalues of $\mathbf{R}(T_A)$. Since $\mathbf{R}(T_A)$ is semi positive defined, $\sigma_i \geq 0$, $i = 1, \dots, n$. And because $\dim(\mathbf{R}(T_A)) \geq 2$, $\sigma_1 \geq \sigma_2 > 0$. Perform eigenvalue decomposition on $\mathbf{R}(T_A)$, we have

$$\begin{aligned} \mathbf{R}(T_A) &= \mathbf{U} \mathbf{\Sigma} \mathbf{U}^H \\ \mathbf{\Sigma} &= \text{diag}(\sigma_1, \sigma_2, \dots, \sigma_n) \end{aligned} \quad (68)$$

where $\mathbf{U}$ is a unitary matrix. Then, there is

$$\begin{aligned} \text{tr}(\mathbf{R}(T_A)) \mathbf{I} - \mathbf{R}(T_A) &= \mathbf{U} \mathbf{\Sigma}' \mathbf{U}^H \\ \mathbf{\Sigma}' &= \text{diag} \left(\sum_{i \neq 1} \sigma_i, \sum_{i \neq 2} \sigma_i, \dots, \sum_{i \neq n} \sigma_i \right) \end{aligned} \quad (69)$$

i.e., the eigenvalues of $\text{tr}(\mathbf{R}(T_A)) \mathbf{I} - \mathbf{R}(T_A)$ are $\sum_{i \neq j} \sigma_i$, $j = 1, \dots, n$. Due to the fact that $\sum_{i \neq j} \sigma_i > 0$, $\text{tr}(\mathbf{R}(T_A)) \mathbf{I} - \mathbf{R}(T_A)$ is nonsingular.

APPENDIX B

In this section, we prove that when $\mathbf{R}(T_B) = \mathbf{I}$, the m -D MCSR subspace is the space spanned by the eigenvectors corresponding to first m largest eigenvectors of $\mathbf{R}(T_A)$. When $\mathbf{R}(T_B) = \mathbf{I}$, there is

$$r_{\min} = \min_{\dim(P)=m} \frac{m}{\text{tr}[\mathbf{P} \mathbf{R}(T_A)]}. \quad (70)$$

For convenient, we denote $\mathbf{R}(T_A)$ as $\mathbf{R}_T$ here. We first find the MCSR subspace

$$S_{\min} = \arg \max_{S \in \mathbf{Gr}(m, \mathbb{C}^n)} \text{tr}(\mathbf{P} \mathbf{R}_T). \quad (71)$$

Let $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_n$ be the eigenvalues of $\mathbf{R}_T$ and $v_1, v_2, \dots, v_n$ are there corresponding eigenvectors. Let $S_1 = \text{span}\{v_1, v_2, \dots, v_m\}$. For any m -D subspace $S_2 \neq S_1$, let $E_0 = S_1 \cap S_2$ and suppose that $S_1 = E_0 \oplus E_1$, $S_2 = E_0 \oplus E_2$. Then $E_1 \subset S_1$, $E_2 \subset S_1^\perp$, that is,

$$\begin{aligned} E_1 &\subset \text{span}\{v_1, v_2, \dots, v_m\} \\ E_2 &\subset \text{span}\{v_{m+1}, v_{m+2}, \dots, v_n\}. \end{aligned} \quad (72)$$

Assume that $\dim(E_1) = \dim(E_2) = k$. According to the min-max theorem, there is

$$\begin{aligned} \text{tr}(\mathbf{P}_{E_1} \mathbf{R}_T) &\geq k\lambda_m \\ \text{tr}(\mathbf{P}_{E_2} \mathbf{R}_T) &\leq k\lambda_{m+1}. \end{aligned} \quad (73)$$

Hence,

$$\begin{aligned} \text{tr}(\mathbf{P}_{S_1} \mathbf{R}_T) &= \text{tr}(\mathbf{P}_{E_0} \mathbf{R}_T) + \text{tr}(\mathbf{P}_{E_1} \mathbf{R}_T) \\ &\geq \text{tr}(\mathbf{P}_{E_0} \mathbf{R}_T) + \text{tr}(\mathbf{P}_{E_2} \mathbf{R}_T) = \text{tr}(\mathbf{P}_{W_2} \mathbf{R}_T) \end{aligned} \quad (74)$$

Hence, one of the MCSR subspace is

$$S_{\min} = \text{span}\{v_1, v_2, \dots, v_m\}$$

and its corresponding orthogonal projecting matrix of $S_{\min}$ is

$$\begin{aligned} \mathbf{P}_{\min} &= \mathbf{V}_{1:m} \mathbf{V}_{1:m}^H \\ \mathbf{V}_{1:m} &= [v_1 \quad v_2 \quad \dots \quad v_m]. \end{aligned} \quad (75)$$

Substitute (75) into (70), we have the MCSR $r_{\min} = m / \sum_{i=1}^m \sigma_i$. Similar to what we have discussed in Section II-B, if $\sigma_m \neq \sigma_{m+1}$, it has the unique solution as shown in (70). If $\sigma_m = \sigma_{m+1}$, $\mathbf{P}_{\min}$ has multiple solutions. Suppose $\sigma_k = \sigma_m$, $k = m_1, m_1 + 1, \dots, m_2$, where $m_1 \leq m < m_2$. Then, the eigenvectors $v_{m_1}, \dots, v_{m_2}$ in (78) can be replaced by any standard orthogonal base of $\text{span}\{v_{m_1}, \dots, v_{m_2}\}$. Hence, $\mathbf{P}_{\min}$ has multiple solutions and can be written as

$$\begin{aligned} \mathbf{P}_{\min} &= \mathbf{P}_{W_1} + \mathbf{P}_{W_2} \\ W_1 &= \text{span}\{v_1, v_2, \dots, v_{m_1-1}\} \\ W_2 &\subset \text{span}\{v_{m_1}, v_{m_1+1}, \dots, v_{m_2}\} \end{aligned} \quad (76)$$

where $\mathbf{P}_{W_1}$, $\mathbf{P}_{W_2}$ is the orthogonal projection matrices of the subspaces W_1 and W_2 , respectively.

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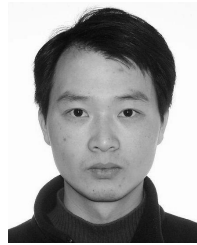
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